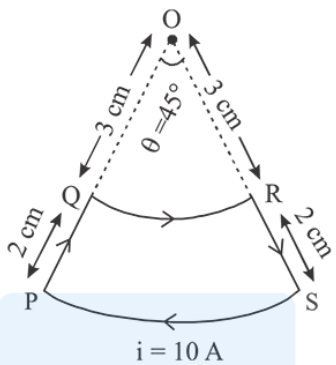
JEE Mains 2019 Chapter wise Question Bank

Magnetic Effects of Current - Questions

Q1

A current loop, having two circular arcs joined by two radial lines is shown in the figure. It carries a current of 10 A. The magnetic field at point O will be close to:



(1) 1.0×10^{-7} T (2) 1.5×10^{-7} T
(3) 1.5×10^{-5} T (4) 1.0×10^{-5} T

9 Jan Morning

Q2

One of the two identical conducting wires of length L is bent in the form of a circular loop and the other one into a circular coil of N identical turns. If the same current is passed in both, the ratio of the magnetic field at the central of the loop (B_1) to that at the centre of the coil (B_C), i.e, $\frac{B_L}{B_C}$ will be:

(1) N (2) $\frac{1}{N}$

(3) N^2 (4) $\frac{1}{N^2}$

9 Jan Evening

Q3

A particle having the same charge as of electron moves in a circular path of radius 0.5 cm under the influence of a magnetic field of 0.5T. If an electric field of 100V/m makes it to move in a straight path then the mass of the particle is (Given charge of electron = 1.6×10^{-19}C)

(1) 9.1×10^{-31} kg (2) 1.6×10^{-27} kg
(3) 1.6×10^{-19} kg (4) 2.0×10^{-24} kg

9 Jan Evening

Q4

In an experiment, electrons are accelerated, from rest, by applying a voltage of 500 V. Calculate the radius of the path if a magnetic field 100 mT is then applied.

[Charge of the electron = 1.6×10^{-19} C
Mass of the electron = 9.1×10^{-31} kg]

(1) 7.5×10^{-3} m (2) 7.5×10^{-2} m
(3) 7.5 m (4) 7.5×10^{-4} m

11 Jan Morning

Q5

The region between $y=0$ and $y=d$ contains a magnetic field $\vec{B}=B\hat{z}$. A particle of mass m and charge q enters the region with a velocity $\vec{v}=v\hat{i}$. if $d=\frac{mv}{2qB}$, the acceleration of the charged particle at the point of its emergence at the other side is :

(1) $\frac{qvB}{m}\left(\frac{1}{2}\hat{i}-\frac{\sqrt{3}}{2}\hat{j}\right)$

(2) $\frac{qvB}{m}\left(\frac{\sqrt{3}}{2}\hat{i}+\frac{1}{2}\hat{j}\right)$

(3) $\frac{qvB}{m}\left(\frac{-\hat{j}+\hat{i}}{\sqrt{2}}\right)$

(4) $\frac{qvB}{m}\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

11 Jan Evening

Q6

A proton and an α-particle (with their masses in the ratio of 1 : 4 and charges in the ratio 1 : 2) are accelerated from rest through a potential difference V. If a uniform magnetic field (B) is set up perpendicular to their velocities, the ratio of the radii $r_p : r_\alpha$ of the circular paths descrfibed by them will be:

(1) $1:\sqrt{2}$ (2) $1:2$

(3) $1:3$ (4) $1:\sqrt{3}$

12 Jan Morning

Q7

A circular coil having N turns and radius r carries a current I. It is held in the XZ plane in a magnetic field $B\hat{i}$. The torque on the coil due to the magnetic field is :

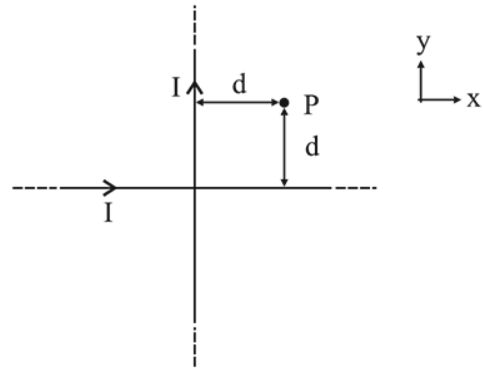
(1) $\frac{Br^2I}{\pi N}$ (2) $B\pi r^2 I N$

(3) $\frac{B\pi r^2 I}{N}$ (4) Zero

8 April Morning

Q8

Two very long, straight, and insulated wires are kept at 90° angle from each other in xy-plane as shown in the figure.



These wires carry currents of equal magnitude I, whose directions are shown in the figure. The net magnetic field at point P will be :

(1) Zero (2) $-\frac{\mu_0 I}{2\pi d}(\hat{x}+\hat{y})$

(3) $\frac{+\mu_0 I}{\pi d}(\hat{z})$ (4) $\frac{\mu_0 I}{2\pi d}(\hat{x}+\hat{y})$

8 April Evening

Q9

A rectangular coil (Dimension 5 cm × 2.5 cm) with 100 turns, carrying a current of 3 A in the clock-wise direction, is kept centered at the origin and in the X-Z plane. A magnetic field of 1 T is applied along X-axis. If the coil is tilted through 45° about Z-axis, then the torque on the coil is:

(1) 0.38 Nm (2) 0.55 Nm

(3) 0.42 Nm (4) 0.27 Nm

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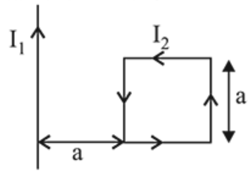
Q10

A rigid square of loop of side 'a' and carrying current I_2 is lying on a horizontal surface near a long current I_1 carrying wire in the same plane as shown in figure. The net force on the loop due to the wire will be:



(1) Repulsive and equal to $\frac{\mu_o I_1 I_2}{2\pi}$

(2) Attractive and equal to $\frac{\mu_o I_1 I_2}{3\pi}$

(3) Repulsive and equal to $\frac{\mu_o I_1 I_2}{4\pi}$

(4) Zero

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Q11

A moving coil galvanometer has a coil with 175 turns and area 1 cm^2. It uses a torsion band of torsion constant 10^{-6} N-m/rad. The coil is placed in a magnetic field B parallel to its plane. The coil deflects by 1° for a current of 1mA. The value of B (in Tesla) is approximately:

(1) 10^{-4} (2) 10^{-2} (3) 10^{-1} (4) 10^{-3}

9 April Evening

Q12

A moving coil galvanometer allows a full scale current of 10^{-4} A. A series resistance of 2 MΩ is required to convert the above galvanometer into a voltmeter of range 0 – 5 V. Therefore the value of shunt resistance required to convert the above galvanometer into an ammeter of range 0 – 10 mA is :

(1) 500 Ω (2) 100 Ω (3) 200 Ω (4) 10 Ω

10 April Morning

Q13

A proton, an electron, and a Helium nucleus, have the same energy. They are in circular orbits in a plane due to magnetic field perpendicular to the plane. Let r_p, r_e and r_{He} be their respective radii, then,

(1) $r_e > r_p = r_{He}$ (2) $r_e < r_p = r_{He}$

(3) $r_e < r_p < r_{He}$ (4) $r_e > r_p > r_{He}$

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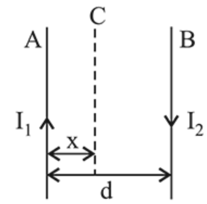
Q14

Two wires A & B are carrying currents I_1 and I_2 as shown in the figure. The separation between them is d. A third wire C carrying a current I is to be kept parallel to them at a distance x from A such that the net force acting on it is zero. The possible values of x are :



(1) $x = \left(\frac{I_1}{I_1 - I_2}\right)d$ and $x = \frac{I_2}{(I_1 + I_2)}d$

(2) $x = \left(\frac{I_2}{(I_1 + I_2)}\right)d$ and $x = \left(\frac{I_2}{(I_1 - I_2)}\right)d$

(3) $x = \left(\frac{I_1}{(I_1 + I_2)}\right)d$ and $x = \left(\frac{I_2}{(I_1 - I_2)}\right)d$

(4) $x = \pm\frac{I_1 d}{(I_1 - I_2)}$

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Q15

The magnitude of the magnetic field at the center of an equilateral triangular loop of side 1 m which is carrying a current of 10 A is :

[Take $\mu_o = 4\pi \times 10^{-7}$ NA^{-2}]

(1) 18 μT (2) 9 μT (3) 3 μT (4) 1 μT

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Q16

A square loop is carrying a steady current I and the magnitude of its magnetic dipole moment is m. If this square loop is changed to a circular loop and it carries the same current, the magnitude of the magnetic dipole moment of circular loop will be :

(1) $\frac{m}{\neq}$ (2) $\frac{3m}{\neq}$ (3) $\frac{2m}{\neq}$ (4) $\frac{4m}{\neq}$

10 April Evening

Q17

A thin ring of 10 cm radius carries a uniformly distributed charge. The ring rotates at a constant angular speed of $40\,\pi$ rad s^{-1} about its axis, perpendicular to its plane. If the magnetic field at its centre is 3.8×10^{-9} T, then the charge carried by the ring is close to ($\mu_0 = 4\pi \times 10^{-7}$ N/A^2).

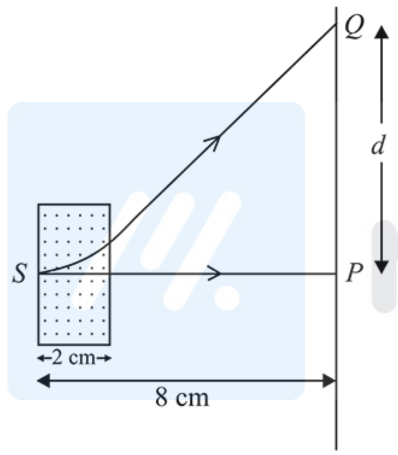
(1) 2×10^{-6}C (2) 3×10^{-5}C (3) 4×10^{-5}C (4) 7×10^{-6}C

12 April Morning

Q18

An electron, moving along the x-axis with an initial energy of 100 eV, enters a region of magnetic field $\vec{B} = (1.5\times10^{-3}\text{T})\hat{k}$ at S (see figure). The field extends between $x = 0$ and $x = 2$ cm. The electron is detected at the point Q on a screen placed 8 cm away from the point S. The distance d between P and Q (on the screen) is :

(Electron's charge $= 1.6 \times 10^{-19}$ C, mass of electron $= 9.1 \times 10^{-31}$ kg)



(1) 11.65 cm (2) 12.87 cm

(3) 1.22 cm (4) 2.25 cm

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Magnetic Effects of Current - Answers

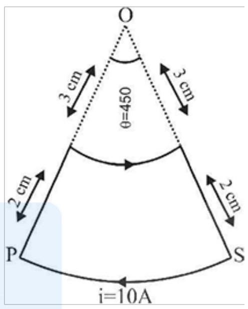
Q1

(4)

There will be no magnetic field at O due to wire PQ and RS

Magnetic field at 'O' due to arc QR

$$= \frac{\mu_0}{4\pi} \frac{\left(\frac{\pi}{4}\right).I}{r_1}$$



Magnetic field at 'O' due to are PS

$$= \frac{\mu_0}{4\pi} \frac{\left(\frac{\pi}{4}\right).I}{r_2}$$

$\therefore$ Net magnetic field at 'O'

$$B = \frac{\mu_0}{4\pi}(\pi/4)\times 10\left[\frac{1}{(3\times10^{-2})} - \frac{1}{(5\times10^{-2})}\right]$$

$$\Rightarrow |\vec{B}| = \frac{\pi}{3}\times10^{-5}\,T \approx 1\times10^{-5}\,T$$

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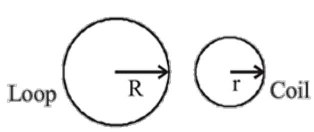
Q2

(4)



$L = 2\pi R \quad L = N\times 2\pi r$

$R = Nr \Rightarrow r = \frac{R}{N}$

$B_{Loop} = \frac{\mu_0 i}{2R} \quad B_{coil} = \frac{\mu_0 Ni}{2r} = \frac{\mu_0 Ni}{2\left(\frac{R}{N}\right)} = \frac{\mu_0 N^2 i}{2R}$

$\therefore \frac{B_L}{B_C} = \frac{1}{N^2}$

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Q3

(4) As particle is moving along a circular path

$\therefore R = \frac{mv}{qB}$ ...(i)

Path is straight line, then

$qE = qvB$

$E = vB \Rightarrow v = \frac{E}{B}$ ....(ii)

From equation (i) and (ii)

$$m = \frac{qB^2R}{E} = \frac{1.6\times10^{-19}\times(0.5)^2\times0.5\times10^{-2}}{100}$$

$\therefore m = 2.0\times10^{-24}\,kg$

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Q4

(4) Radius of the path (r) is given by $r = \frac{mv}{qB}$

$$r = \frac{\sqrt{2mk}}{eB} \quad \left(\because\ p = mv = \sqrt{2mk}\right)$$

$$= \frac{\sqrt{2meV}}{eB} \quad (\because\ k = eV)$$

$$r = \frac{\sqrt{\frac{2m}{e}V}}{B} = \frac{\sqrt{\frac{2\times9.1\times10^{-31}}{1.6\times10^{-19}}(500)}}{100\times10^{-3}}$$

$$r = \frac{\sqrt{\frac{9.1}{0.16}\times10^{-10}}}{10^{-1}} = \frac{3}{.4}\times10^{-4}$$

$$= 7.5\times10^{-4}$$

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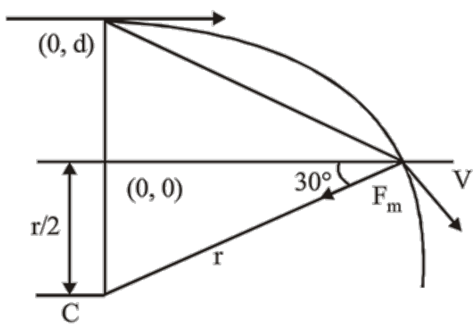
Q5

(BONUS)

Assuming particle enters from (0, d)

$$r = \frac{mv}{qB}, \qquad d = \frac{r}{2}$$



$$a = \frac{qVB}{m}\left[\frac{-\sqrt{3}i - j}{2}\right]$$

this option is not given in the all above four choices.

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Q6

(1) Radius of the circular path will be $r = \dfrac{mv}{qB}$

$$\Rightarrow r = \frac{\sqrt{2mKE}}{qB} \quad (\because\ p = mv = \sqrt{2mKE})$$

$\because$ $KE = q\Delta V$

$$\therefore\ r = \frac{\sqrt{2mq\Delta V}}{qB} \Rightarrow r \propto \sqrt{\frac{m}{q}}$$

$$\therefore\ \frac{r_p}{r_\alpha} = \frac{1}{\sqrt{2}}$$

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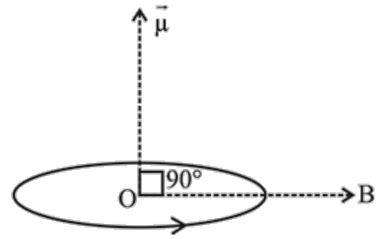
Q7

(2) $|\vec{\tau}| = |\vec{\mu} \times \vec{B}|$ $\qquad [\mu = NIA]$

$= NIA \times B \sin 90^o$ $\qquad [A = \pi r^2]$

$\Rightarrow \tau = NI\pi r^2 B$



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Q8

(1) $\vec{B} = \vec{B_1} + \vec{B_2}$

$$= \frac{\mu_0}{2\pi}\cdot\left(\frac{i^o}{d}.\hat{k} + \frac{i^o}{d}(-\hat{k})\right) = 0$$

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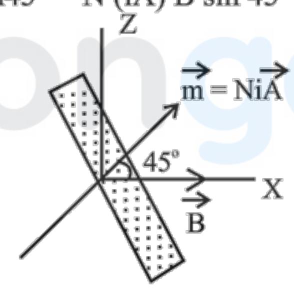
Q9

(4) $\tau = mB \sin 45° = N\,(iA)\,B \sin 45°$



$$= 100 \times 3(5 \times 2.5) \times 10^{-4} \times 1 \times \frac{1}{\sqrt{2}} = 0.27\ \text{Nm}$$

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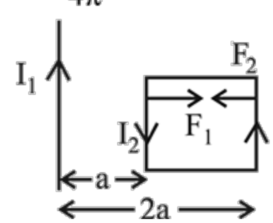
Q10

(3) $F = F_1 - F_2 = \dfrac{\mu_0}{2\pi}\left(\dfrac{I_1 I_2}{a} - \dfrac{I_1 I_2}{2a}\right) \times a$

$$= \frac{\mu_0 i_1 i_2}{4\pi} \text{ (Repulsive)}$$



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Q11

(4) $C\theta = NBiA \sin 90°$

or $10^{-6}\left(\frac{\pi}{180}\right) = 175B(10^{-3}) \times 10^{-4}$

$\therefore B = 10^{-3}$ T

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Q12

(Bonus) $v = i_g (R + G)$

$\Rightarrow 5 = 10^{-4} (2 \times 10^6 + x)$

$x = -195 \times 10^4 \Omega$

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Q13

(2) As $mvr = qvB \Rightarrow r = \frac{mv}{qB} = \frac{\sqrt{2mK.E.}}{qB}$

[As : $\frac{1}{2}mv^2 = K.E.$

$\Rightarrow m^2v^2 = 2mK.E.$

$\Rightarrow mv = \sqrt{2mK.E.}$]

For proton, electron and α-particle,
$m_{He} = 4m_p$ and $m_p >> m_e$
Also $a_{He} = 2q_p$ and $q_p = q_e$
$\therefore$ As KE of all the particles is same then,

$r \propto \frac{\sqrt{m}}{q}$

$\therefore r_{He} = r_p > r_e$

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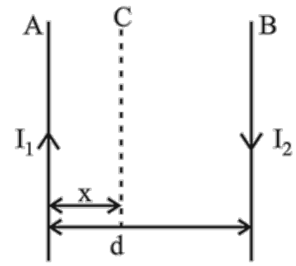
Q14

(4)



As net force on the third wire C is zero.

$\Rightarrow \vec{F} = \frac{_0I_1}{2\pi x} + \frac{_0I_2}{2\pi(d-x)} = 0$

$\frac{_0I_1}{2\pi x} = \frac{_0I_2}{2\pi(x-d)}$

$I_1x - I_1d = I_2x$

$x = \frac{I_1 d}{I_1 - I_2}$

Two cases may be possible if $I_1 > I_2$ or $I_2 > I_1$

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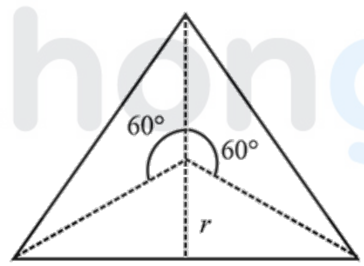
Q15

(1)



$r = \left(\frac{1}{3}\right)(a \sin 60)$

$r = \frac{a}{3} \times \frac{\sqrt{3}}{2} = \left(\frac{a}{2\sqrt{3}}\right)$

$B_0 = 3\left[\frac{\mu_0 I}{4\pi r}(\sin 60 + \sin 60)\right]$

$= \frac{3\mu_0 I}{4\pi\left(\frac{a}{2\sqrt{3}}\right)} \times (2)\left(\frac{\sqrt{3}}{2}\right)$

$= \frac{9}{2}\left(\frac{\mu_0 I}{\pi a}\right)$

$= \frac{9 \times 2 \times 10^{-7} \times 10}{1}$

$= 18\ \mu T$

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Q16

(4) Let a be the area of the square and r be the radius of circular loop.

$$2\pi r = 4a \Rightarrow r = \left(\frac{2a}{\pi}\right)$$

For square

$M = (I)\,a^2$

For circular loop

$M_1 = (I)\pi r^2$

$$M_1 = (I)(\pi)\left(\frac{4a^2}{\pi^2}\right)$$

$$M_1 = \frac{4Ia^2}{\pi}$$

$$M_1 = \frac{4M}{\pi} \qquad (\because M = Ia^2)$$

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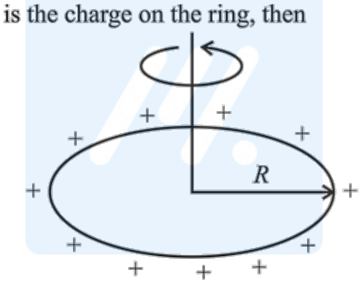
Q17

(2) If q is the charge on the ring, then



$$i = \frac{q}{T} = \frac{q\omega}{2\pi}$$

Magnetic field,

$$B = \frac{\mu_0 i}{2R}$$

$$= \frac{\mu_0\left(\frac{q\omega}{2\pi}\right)}{2R}$$

or $\quad 3.8 \times 10^{-9} = \left(\frac{\mu_0}{4\pi}\right)\frac{q\omega}{R} = \left(10^{-7}\right)\frac{q \times 40\pi}{0.10}$

$\therefore \quad q = 3 \times 10^{-5}\,\text{C}.$

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Q18

(2)

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